

Practice Problem Solutions

Data Analysis & Outlier Detection

Problem 1: Z-Score Method (Normal Data)

Dataset: Age of students in a class (n = 20)

Calculations

- **Mean (μ):** 28.40
- **Standard Deviation (σ):** 12.13 (Sample Standard Deviation)

Using the Z-score formula $Z = (X - \mu) / \sigma$, we check for values where $|Z| > 3$.

- For X = 70: $Z = (70 - 28.40) / 11.96 = 3.43$

Answers:

- **Outliers Identified:** [70]
- **New Dataset Length:** 19 (After removing the single outlier)

Problem 2: IQR Method (Skewed Data)

Dataset: Monthly salaries in thousand BDT (n = 25)

Calculations

Metric	Value
Q1 (25th Percentile)	35.00
Q3 (75th Percentile)	70.00
IQR (Q3 - Q1)	35.00
Lower Bound (Q1 - 1.5 * IQR)	-17.50
Upper Bound (Q3 + 1.5 * IQR)	122.50

Answers:

- **Outliers Identified:** [150, 200, 250, 500] (Values > 122.50)
- **After Capping to Bounds:** All outliers are replaced with the upper bound (122.50).
- **New Min:** 25.00 | **New Max:** 122.50

Problem 3: Choose the Right Method

Dataset A (Exam Scores)

Histogram Conceptualization: The data has a heavy concentration of high scores (several 100s) but is dragged down by two extremely low scores (15, 20). This represents a **Left-Skewed (Negatively Skewed)** distribution.

Chosen Method: IQR Method. The extreme low values would artificially pull down the mean and inflate the standard deviation, making Z-scores unreliable. IQR is median-based and robust against extreme skewness.

Dataset B (House Prices)

Histogram Conceptualization: The dataset has standard values clustered between 30 and 120 but extends dramatically to 1200. This represents a **Right-Skewed (Positively Skewed)** distribution, typical for real estate.

Chosen Method: IQR Method. For the same reasons as Dataset A, right-skewed data distorts the mean, rendering the Z-score masking effect highly problematic.

Applying IQR to Dataset B:

- $Q1 = 45.00$, $Q3 = 200.00$, $IQR = 155.00$
- Upper Bound = 432.50

Answers:

- **Outliers found in Dataset B:** 4 ([500, 800, 1000, 1200])

Problem 4: Winsorization (Percentile Method)

Dataset: Product ratings out of 10 ($n = 25$)

Calculations

- **5th Percentile:** 1.00
- **95th Percentile:** 10.00

All values below 1.00 are capped to 1.00, and all values above 10.00 are capped to 10.00.

Answers:

- **New Min after Winsorization:** 1.00
- **New Max after Winsorization:** 10.00

Why is Winsorization better here? Product ratings operate on a fixed scale. Extreme values (like 0.5) represent genuine, albeit harsh, user feedback rather than data entry errors. Deleting them ignores valid customer sentiment and artificially reduces the sample size. Winsorization limits the disproportionate impact of extremes while retaining the data points.

Problem 5: Mixed Methods - Compare Z-Score & IQR

Dataset: Highly dispersed product ratings (n = 33)

Histogram Conceptualization: The data has extreme values up to 2000 compared to a median around 38. This is severely **Right-Skewed**.

Method	Outlier Count	Identified Outliers
Z-Score ($ Z > 3$)	1	[2000]
IQR Method	4	[200, 500, 1000, 2000]

Answers:

- **Which method found more?** The **IQR Method** found more outliers (4 vs 1).
- **Why?** The extreme values (500, 1000, 2000) heavily inflated the standard deviation (379.66), making the Z-score threshold so wide that it masked the smaller outliers. This is known as the "masking effect."
- **Which to trust? IQR.** It uses rank-based percentiles which are immune to extreme distortion, successfully identifying all anomalous values in highly skewed data.

Problem 6: Real-World Scenario - Employee Bonus

Dataset: Bonus amounts in thousand BDT (n = 33)

Initial Observations

- **Mean:** 42.24
- **Median:** 12.00
- **Mode(s):** 7, 8, 10

Observation: The mean (42.24) is significantly higher than both the median and the modes. This indicates a strong positive skew caused by executive-level or exceptional bonuses dragging the average upward.

IQR Method Outliers

Using $Q1 = 8.00$, $Q3 = 40.00$, the upper bound is 88.00.

Outliers identified: [100, 120, 150, 200, 300]

Winsorization (x=3%)

- **3rd Percentile:** 5.00
- **97th Percentile:** 204.00

Answers:

- **New Min after Winsorization:** 5.00
- **New Max after Winsorization:** 204.00
- **New Mean:** 39.33

Is the new mean more reasonable? Yes. The original mean of 42.24 was unrepresentative of the typical employee, as half the company received 12.00 or less. The new Winsorized mean (39.33) limits the extreme executive bonuses, resulting in an average that is much closer to a realistic baseline for policy-making.